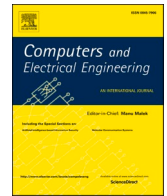




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Personalized device authentication scheme using Q-learning-based decision-making with the aid of transfer fuzzy learning for IIoT devices in zero trust network (PDA-QLTFL)

Anamika Singh^a, Rajesh Kumar Dhanaraj^{b,*}, Anupam Kumar Sharma^a^a School of Computing Science and Engineering, Galgotias University, India^b Symbiosis Institute of Computer Studies and Research (SICSR), Symbiosis International (Deemed University), Pune, India

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ABSTRACT

The IIoT infrastructure builds around devices without human intervention. Likewise, human identification and authentication before accessing the devices, the machines involved in the network also need authentication to eliminate the risk of anomaly. The existing works provided the means for device authentication but failed to focus on the personalized authentication of devices based on their security requirements. The zero-trust network has many devices, and each device before becoming a network member and gaining access to the network needs to be verified. It is a time-consuming process to verify each device and sometimes the same security policy does not meet the security requirements of all the devices. The proposed paper highlighted an approach where each device authenticity is measured individually depending on the associated device security requirements and the profile of the device. The underlying scheme breaks into different phases: making blocks of the device data and assigning hash values to each block to uniquely identify a particular block containing associated device data. Next, each device data is encrypted using the key and then decrypted by the intended receptor holding the key to obtain the original data to ensure the device data's confidentiality and integrity. After that, the profiling of the device is maintained based on the personalized authentication being performed to analyze the access level and the security of the requirements of the devices in the network. The proposed PDA-QLTFL outperforms the existing approaches on the parameters false positive rate, data confidentiality rate, data integrity rate, and network access time by 5.116 %, 5.446 %, 5.82 % and 3.308 % respectively.

1. Introduction

The globe has entered the fourth industrial revolution resulting in a technological transformation. the shifting of the traditional paradigm to the digital platform has experienced remarkable swiftness irrespective of the domain and execution criteria. One of the many contributing technologies that play a pivotal role in the revolution is the Internet of Things which is not restricted to a significant sector. The IIoT (Industrial Internet of Things) is where everything builds around devices. IIoT brings devices to the network making the devices and network more autonomous. IIoT brings together the industrial devices and the network consequence in enhanced and effective industrial productivity. IIoT enables the devices to communicate with their peers and other industrial systems to automate the

* Corresponding author.

E-mail address: sangeraje@gmail.com (R.K. Dhanaraj).<https://doi.org/10.1016/j.compeleceng.2024.109435>

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process remotely. The IIoT presents a vision of smart connected industrial devices operating over the network to establish a smarter enterprise. IIoT being a class of IoT builds a smarter and more powerful framework for industrial infrastructure that incorporates various IoT components including sensors, actuators, wireless devices, high-speed internet connections so, and so forth. IIoT turns into a transformative approach with enhanced productivity and security. IIoT is a network of interconnected devices that collects, enables, and ensures secure data communication IIoT focuses on establishing a secure and robust industrial environment handling large-scale networks seamlessly. There is a list of technologies that act as a backbone for the growth and notable adoption of IIoT infrastructure and the integrating technologies include IoT, cloud computing, artificial intelligence, big data analytics, M2M communications, etc. Despite of all the contributions, the IIoT environment is always prone to cyber-attacks as the infrastructure build and operate on devices.

The zero-trust security model is a framework that doesn't trust users based on their perimeter and asks them to verify every time to access network resources. A zero-trust framework addresses many aspects of the industrial environment, all centering around the concept of never trusting, and always verifying. The traditional security models cannot adapt to the current security requirements of emerging technologies. The zero-trust security model is a new security solution that validates and confirms the authenticity of the device or the resource seeking permission to access the network irrespective of the access call from the internal or external networks. In zero-trust architecture, all the devices and resources are assumed to be untrusted by default and provide a new level of security boundary to the more complex networks. Traditional perimeter-based network security with fixed boundary security solutions based on trust is no longer considered a strong security solution for emerging complex networks. The emergence of zero trust security architecture verifies and confirms the device's authenticity and originality as no device can now be trusted and allowed to access the network simply based on its location and address.

Covering the approach behind the zero trust model, NCSC stated ten core principles of the zero trust security model-1) understanding of the architecture, devices, policies, and services; 2) formulating a single but strong identity; 3) a powerful device analysis; 4) authenticate everywhere; 5) understanding the requirements and security of the devices; 6) surveillance of device and services; 7) policies depending on the utility of devices and services; 8) handling and controlling access to device and services; 9) no trust on the network, including the local network; 10) consider services formulated for zero trust.

In [1], the approach named Consistent Round Hash Optimized Secure Remote Password (CRH-SRP) is discussed that focuses on mutual authentication for establishing an authentic and safe data transfer over the network. Moreover, a sound system was designed with system integrity and stability even when a complex industrial environment is considered. the results of the underlying concept come up with the data transfer with minimum communication overhead and computational cost. Despite the discussed key points of the approach, the CRH-SRP fails to establish data confidentiality and data integrity. To address this aspect, the proposed paper aims to ensure data confidentiality and data integrity with the aid of a hash-function-based substitution-permutation network that establishes sound data security. In the network, all the credentials of the industry are handled digitally and where devices are involved there will be the risk of a data breach. Therefore, the data needs to be safeguarded from being modified by any unauthorized or infected device. Data confidentiality and data integrity are the two aspects that should be addressed and guaranteed in the network of thousands of complex devices. For sound industrial infrastructure, data security is the major consideration that needs to be focused on and maintained without fail.

With the emergence of complex networks in the IIoT environment, security questions are always the major aspect of consideration. The IIoT environment and its functionality build around thousands of devices and associated resources and thus the network is always prone to malicious threats. The identity of the device irrespective of internal or external needs to be maintained to reduce the risk of attack. The zero-trust network provides an ambient solution to complex network security requirements. ZTSM (zero trust security model) works in a way that it first verifies the authenticity of the device without allowing access to the network. The continuous verification of the devices in the network adversely affects the functionality and efficiency of the system model. When any device enters the network, the model validates the authenticity before allowing access. Verifying and authenticating thousands of devices is a time-consuming process that turns into a gap in the ZTS model. Thus, profiling of the device is performed to avoid authenticating the same device each time the device enters the network. As discussed above, the network is always prone to anomalies and device profiling provides an efficient approach for appropriate anomaly detection. The network comprises devices with different configurations handling varying levels of complexity and security requirements. Authentication schemes can be custom-made by deep learning techniques that decide the type of authentication for any given device to access a network resource in question.

The proposed concept discussed in the paper functions in different stages. The "smart setup" begins with storing the individual device data in blocks with each block handling a particular device data. Then the next consideration is to maintain and ensure data confidentiality and data integrity to the data stored in ZTN. Assigning hash values to each identical block and the data stored in the block is encrypted using the symmetric key to establish secure and leakproof data transfer over the network. during the communication, the policy administrator examines the resource access patterns and constructs the device profile based on these patterns for personalized authentication. Profiling of the device is one aspect of personalized authentication, the other factor that contributes to more realistic personalized authentication is the security level a resource possesses for permitting access. the proposed paper adopted a deep learning approach for accurate and optimized decision-making for personalized authentication schemes. deep learning facilitates more appropriate and relevant decision-making without human intervention. for facilitating the enhanced and accurate personalization of devices, the paper focused on transfer fuzzy learning to evaluate the security level of resources being accessed by the devices as not all devices in the network going to access the same level of security resources. Therefore, the classification is performed with the aid of transfer fuzzy learning to assign the values to easily identify the security level and hence provide two-way personal authentication criteria for the devices in the IIoT environment to boost up the working of the zero-trust security model.

The zero-trust model works on continuous authentication but has a drawback when comes to the industrial environment.

Continuous authentication of thousands of devices based on different undefined parameters has adverse effects on the performance of the industries operating digitally. Transfer learning is a deep learning technique that is employed to enhance decision-making by acquiring information from the above-performed phases of evaluating trust and confidentiality and also in terms of cryptographic operation performed as zero trust network treats every device irrespective of being a member of a network or not, as a malicious and untrusted device.

Integrating hash function-based substitution-permutation networks with transfer fuzzy learning will provide ambient solutions to the problems discussed in this section above. With the adopted approach, unnecessary device authentication is aimed to be eliminated and therefore uplifting the production and functioning of the IIoT environment without any hindrance while maintaining the zero-trust security in place. The authentication of network devices is only performed in case of anomalies or simply when the devices have different behavior or patterns than the usual profile. Enabling deep learning in the zero-trust security model will improve the security of the network by providing the same level of security in the meanwhile reducing the burden of repeated authentications. A customized authentication scheme based on the type of device and type of resource to be accessed with different security levels strengthens the security of IIoT in an efficient way.

The major contribution of the research work is as discussed below:

1. Assimilation of deep learning with a zero-trust security model in an IIoT environment provides optimized decision-making without hindering the performance of the network.
2. Eliminates the risk of anomalies in the network with the aid of device profiling.
3. Enabling Transfer fuzzy learning with the aid of pre-trained data empowers the learning in the IIoT environment.
4. The notable performance is achieved in the context of false positive rate, data confidentiality rate, data integrity rate, and network access time evaluated on the simulating dataset and compared to existing methods.

The paper is organized as, [Section 2](#) discusses the related works associated with the adopted methodology, and [Section 3](#) describes the materials and methods used to implement the approach. Next, [Section 4](#) discusses the proposed approach with the system model and related algorithms. [Section 5](#) explains the experimental setup and the result analysis obtained from the performance evaluation. The paper concludes with [Section 6](#), discussing the conclusion and future scope.

2. Related works

With the widespread digitalization in every corner of the globe irrespective of the operating environment, cybersecurity and maintaining workplace decorum is one of the major considerations. To provide ambient solutions the security questions with improved and high-security requirements in operating IIoT environment, the zero-trust security model is employed to provide data confidentiality and data integrity of the devices in the network in practice [2]. Zero trust security performs on the novel approach of not trusting any device either the member of the network or granting access permission as a new one [3]. The verification and authentication of each device every time it enters the network is a tedious and time-consuming task that makes the environment built on devices dry up resulting in weak performance and productivity as well. Zero trust security executes on three different sections- software, hardware, and organizational [4]. In [5] Zero trust aims to minimize cyber risk and enhance the operating experience with sound digital association. The paper discussing the architectural concept model of i-ZTA, which provides data security from untrusted devices with the aid of machine learning is explained in [6]. The introduction of zero trust provides real-time monitoring of system and device security, evaluating the risk of device requests and verifying the authorization based on the risk evaluation [7].

In [8], the emergence and deployment of deep reinforcement learning in data-oriented decision-making in financial organizations is delineated. Moreover, the work focused on reviewing the growth of three broad concepts Deep Learning (DL), Reinforcement Learning (RL), and Deep Reinforcement Learning (DRL). IIoT enables the environment in empowering connectivity among devices globally. An in-depth systematic review to explain the development and employment of IIoT concepts is delineated in [9]. IIoT is an infrastructure that operates on and around devices and has modified the functioning of underlying industries. In [10], the explanation of the is explained with three different concepts of IIoT- the industrial, the internet and the things. IIoT is not a single concept with single operating technology but has number of technologies that incorporate and support the working of IIoT framework is analysed in [11]. The considering technologies are- cloud computing, big data, artificial intelligence, smart devices, wireless sensor network, Bluetooth, ZigBee so and so forth [12]. In [13], the profiling of patterns and the behaviour of the devices and humans is an effective mean to examine the functioning and authenticity of the device and user in the operating network is focused. The emerging industries believe in BYOD (Bring Your Own Device) phenomenon and therefore, device profiling plays pivotal role in examining and identifying unusual user and device behaviour to reduce cyber-crime [14]. A device profiling technique for detecting the threats with the aid of inter-arrival time of the devices for automatized identification of the unusual patterns of the entered devices. An emerging technique termed as radio Frequency (RF) is proposed in [15] for generating modernized device profiling. The work adopted the Short Time Fourier Transform to utilize STFT for empowered RF fingerprinting.

In the IIoT environment, data security and originality is the major consideration to be looked upon. The blended approach that provides an ambient solution to security issues is proposed in [16] that offers core authentication challenges solution- data integrity and restricted resource access. smartphones nowadays, becoming smarter enables users to store their credentials in one place. But on the other hand, permanent login to the device makes the device and credential data more prone to attacks. The underlying problem is considered in [17], and the proposed paper brought an approach of continuous authentication via a deep learning approach known as DeepAuthen. Machine learning and deep learning approaches are the most convenient way of ensuring security in IoT environments

including IoMT as well [18]. A systematic review is made in [19] to explore various techniques involved in generating IoT device profiling and incorporating vulnerabilities thereafter. Transfer learning is a sub-category of machine learning that witnessed sound growth in recent years. Irrespective of the conventional approaches of machine learning, transfer learning embraces the pre-trained information or learning facilitating speedy result delivery and accuracy as a consequence is delineated in [20]. Transfer of knowledge between the states or tasks provides a better way of collecting data. Deep learning is always been a convenient approach and has attained notable acceptance worldwide irrespective of the domain the system or the industry is operating upon which offers the opportunity to enhance productivity and fill the gap reported in reinforcement learning [21]. Moreover, the use of fuzzy learning is made in IoT devices to empower smart homes to provide leverage and easy access to blind people for wayfinding [22]. In [23], the protection and guaranteed security of the IoT system-based zero trust security framework is delineated that offers a sound security solution. The various challenges related to authentication, access control techniques, risk management evaluation, etc. are discussed in [24]. Contrary to an existing perimeter-based architecture where the devices are trusted based on some perimeters, zero-trust architecture assumes every device is an untrusted party and verifies the request before permitting access [25]. The implementation of q-learning in the assembly issue in any underlying entity with the objective of finding and providing efficacy in the development of the undertaken assembly execution time is formulated in [26]. Therefore, the q-learning approach is applied to constructing a q-matrix with the aid of observed and calculated q-values resulting from the communication with the operating system that helps in recommending an assembly chronological result [27]. In [28], the application of q-learning in the field of unmanned aerial vehicles is delineated explaining the use and sound optimization in choosing the most appropriate and relevant action or decision to be taken based on the states of the environment depending on the pre-defined policies. In [29], the strategies and notions are retrieved from reinforcement learning, where q-learning is the most admired traditional reinforcement algorithm. A novel approach to adopting an environment with deep-quality learning for e-learning and distance learning concepts is recommended in [30]. A contribution to elaborate on various machine learning algorithms as well as their application and acceptance in different areas of interest is illustrated in [31].

The automated generation of mobile speed profiling of the users of the mobile depending on the deep learning is delineated in [32]. The results produced facilitates high accuracy on the datasets gathered from real environment on 5 G networks. the approach focuses on profiling the speed by keeping track on the day-to-day behaviour and range of speed of the current user. A systematic review is made in [33] highlighting various machine learning algorithm sin practice with their associated roles and performance in real world scenarios. Data collection from diverse sources and managing that data for filtering and retraining the irrelevant data from the gathered dataset is a challenging task in machine learning algorithms for better accuracy and efficient performance of the undertaken algorithm [34]. The applicability and deployment of deep learning algorithms, reinforcement learning algorithms and integrated deep reinforcement learning algorithm-based decision-making in finance industries is reviewed in [35]. The parameters focused in context of the performance of deep learning, deep reinforcement learning are the optimized functioning of the operations, cost-effectiveness and increased market size etc. is mentioned in [36]. In [37], study found that the RL, DL and DRL amplifies the performance and accuracy when compared to conventional algorithms in terms of economic factors, risk management and dealing with ambiguity. The novel approach for the deployment of artificial intelligence in medical sector is explained in [38]. The study made fills the gap related to the data confidentiality and the process of collecting the data form the intended source. In [39], the implementation of the five key principles of zero trust security in institutions of higher learning and businesses is recommended to provide a solution for the identification and security of the device. a survey for introducing the zero-trust security model in device recognition, authentication, controlling user access and the risk assessments is explained in [40]. In [41], multi factor authentication and monitored authentication is recommended as more secure techniques in complex networks as single factor authentication is now easy for intruders to break. The elaborated examination and summarized characteristics are noted in [42] focusing on the current position of the zero-trust security to grasp the complete status of the ZTS. The implementation of deep learning in blockchain technology is proposed in [43] for improving the security level by handling the malicious activities in consensus process of the blockchain architecture which was not focused in existing works. For amplifying the security levels, the novel approach Dueling double deep Q-network with prioritized experience replay was introduced in [44]. A systematic review for application of reinforcement learning in mobile edge computing is performed in [45] facilitating classified services and compelling channels. Different methods for accurate data collection to and identify and handle the challenging situations in data collection process is monitored and solution proposals are modelled in [46].

The challenges associated with the cloud-based systems is explained in [47] moreover focusing on the factors leading to the applicability of ZTS model. In [48], the cost effectiveness and the deployment of zero trust to behave as a defender of various cyber-attacks is discussed. The AI based automated sensing and smart decision-making for IoT environment is proposed in [49] with diverse operating environment. the state-of-the art review was made in [50] in the matter of frameworks, undertaken needs of the system and incorporating technologies. Deep earning, big data and the edge computing are recommended as the core contributing technologies when comes to the future of IIoT for enhanced security and scalability [51]. In [52], an intrusion detection system was made with the aid of machine learning algorithm to identify the attacks in IoT environment. with benchmarks in terms of accuracy by 99.9 %. The reinforcement learning driven charging strategies for the electronic vehicle, the underlying benefactions and target areas are summarized in [53]. In [54], the concept of reinforcement learning, deep learning, q-learning and deep q-learning is explored and comparative explanation is portrayed. In [55], a Takagi-Sugeno fuzzy sliding mode controller approach is adopted to synchronous the distinct and disordered fractional orders moreover adjusts the complicated power grid systems. The implementation of T-S fuzzy method is proposed to adjust the two distinct chaotic and make the magnet synchronous motor sable [56].

The studies further discussed contributed in some way or the other to enhance the performance and efficiency of IIoT environment over zero trust network. despite several improvements, the researchers further lacked in focusing the challenges in the zero-trust architecture. The gaps in the existing approaches are brought up with a solution in the form of the adopted approach PDA-QLTFL.

The personalized authentication based on the device profile and the security level of the resources to be accessed brings a positive transformation to the zero-trust network-aided IIoT environment eliminating the continuous verification. The parameters that were lacking in previous studies are data confidentiality rate, data integrity rate, false positive rate, and network access time which are explained and focused on in the presented paper PDA-QLTFL.

3. Materials and methods

3.1. Materials

The dataset used for simulation in the proposed approach is retrieved from the X-IIoTID DATASET that is maintained by the Muna Al-Hawawreh at the University of New South Wales (UNSW) in Canberra for evaluating the performance of the adopted approach. The dataset comprises 421,417 usual records, 399,417 falsified records, and 59 attributes of the devices and the corresponding resources. The other dataset under consideration is the CIC IoT dataset 20,22¹. The dataset involved is extracted using the following links: <https://www.kaggle.com/datasets/munaalhawawreh/xiiotid-iiot-intrusion-dataset> and <https://www.unb.ca/cic/datasets/iotdataset-2022.html>.

4. Methods

The peer approaches that are referred to in the proposed paper are the CRH-SRP and Cyber Security via Determinism. In [1], data security and steadiness are delineated in straightforward as well as intricate networks. Moreover, the method focused on seamless communication between the devices with the aid of interactive authentication. The proposed methodology uses employs different methods that are collaborated to turn out in a single novel approach. to provide hash values for data and network security, compression-based substitution-permutation is applied. Transfer fuzzy learning aided security level is measured for effective device profiling. Deep learning is incorporated to facilitate the decision personalized decision making for the device to access the network.

5. Proposed methodology

5.1. System model

The approach adopted in the paper works on different phases to accomplish different associated tasks involved in achieving a more efficient ZTS model in the IIoT environment. The basic model for the underlying concept of personalized device authentication is elaborated below in Fig. 1. The overall basic architecture for the personalized device authentication based on the profile and the level of security resource possess for being accessed is breakdown into different steps involving assigning a hash value to each block holding the device data. Next is to perform encryption and decryption on the data to ensure data confidentiality and integrity in order to establish leakproof data transfer among devices over the network. then profiling the devices based on their communication patterns. Lastly, implementing personalized device authentication based on Q-learning with the aid of transfer fuzzy learning to classify various security levels possessed by the resources being accessed by the devices in the network. the model of the PDA-QL is formulated in Fig. 1:

The Fig. 1 demonstrates the approach adopted. PDA-QLTFL foremost stores a particular device's data on an individual block and each block is assigned a hash value to maintain uniqueness among the blocks storing the device's data individually. Each block is then assigned a value that will uniquely identify a block among the thousands over the network and these values are called hash values. For evaluating and assigning hash value, the underlying scheme implements with the aid of the Miyaguchi–Preneel compression function which is an extended class of the Matyas-Meyer-Oseas compression function. After assigning hash values, the device data is secured by encryption-decryption to maintain data confidentiality and data integrity by performing substitution-permutation of bits on the device data. Then comes the personalized authentication that depends on two contributing factors which are the profile of the device and the security level of resources being accessed by the devices in the network. the device profile is created by examining the communication

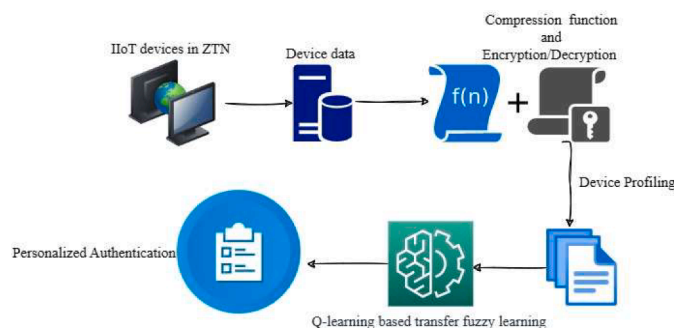


Fig. 1. The overall working model of the proposed (PDA-QLTFL).

pattern. And based on these profiles the q-learning-based personalized authentication is performed to reduce the continuous monitoring and verification overhead of the zero-trust network.

5.2. Miyaguchi–Preneel compression function-based network with the aid of substitution-permutation for device profiling (MPCSP-DP)

The networks are built around thousands of devices and resources that need access to the network. the devices in the network hold immense of confidential information that needs to be kept safe from being accessed by any unauthorized or malicious device present in the network. therefore, to safeguard the confidential information held by the devices, data confidentiality and data integrity are the key considerations to eliminate the risk of falsely modifying the data and adversely affecting the originality of the data relevant to the sound and safe IIoT environment. Therefore, to ensure data confidentiality and integrity, a novel approach Miyaguchi–Preneel compression function is used that assigns unique hash values to each device data block to provide a trustworthy IIoT environment. compression function provided enhanced security and privacy by encoding the plaintext with the random hash values which are only decrypted by the intended device holding the key. Miyaguchi–Preneel compression function is an upgraded version of Matyas–Meyer Oseas compression function and is considered as a single-block-length one way compression function. The compression function works with the aid of device data that is received as the input that is to be encrypted. The structure of the Miyaguchi–Preneel compression function-based network with the aid of substitution-permutation for device profiling (MPCSP-DP) is shown in Fig. 2.

As illustrated in the Fig. 2, the foremost is to formulate a vector matrix that holds the input based on the dataset involved in the simulation process and the features of the devices. Then, on the vector matrix, the underlying compression function is implemented to evaluate the hash value to produce cipher text for the original data, thus ensuring data integrity and confidentiality. The vector matrix is designed with device data and the simulating X-IIoTID sample dataset as follows in Eq. (1):

$$VM = \begin{pmatrix} \alpha_1\beta_1 & \alpha_1\beta_2 & \dots & \alpha_1\beta_x \\ \alpha_2\beta_1 & \alpha_2\beta_2 & \dots & \alpha_2\beta_x \\ \dots & \dots & \dots & \dots \\ \alpha_y\beta_1 & \alpha_y\beta_2 & \dots & \alpha_y\beta_x \end{pmatrix} \quad (1)$$

The vector matrix illustrated in Eq. (1) contains the sequence of device data and the sample dataset applied on all the involved devices. The matrix contains a set of sample data $\alpha = \{\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_y\}$ and the device data $= \{\beta_1, \beta_2, \beta_3, \dots, \beta_x\}$. In the proposed paper, the one-way compression function Miyaguchi–Preneel is used for obtaining the cipher text for the associated plain text.

Plain text is the data handled and stored by a particular device for communication. The cipher text is obtained with the aid of the Miyaguchi–Preneel compression function and its application to the substitution-permutation sequence of bits. The resulting cipher text obtained in the form of hash values is then stored by the policy engine in the form of network access policy. The cipher text is then transferred to the intended receptor to perform the decryption function to acquire associated plain text using the symmetric key that in turns examines the communication pattern among the devices in the network. thus, ensuring the data confidentiality and data integrity by blocking unauthorized access and safeguarding the network from cyber-attack.

The hash value for the data is performed using the adopted compression function. The hash value is a unique value assigned to the original data to create a new arrangement of bits to maintain the privacy of the data. Hash value empowers the authentication process by providing ease to extract the plaintext from the retrieved hashed values using the associated key which is held by the intended receptor only. Only the valid receptor is able to obtain the sent plaintext from the hashed text. The hash values are considered secure as if a small amount of change is made to the data either intentionally or unintentionally, the whole sequence of hashed value gets changed.

Let us assume the block length and the key length of the block be represented as “ B_l ” and “ K_l ” respectively. The encryption that is to be performed on the device data using the symmetric key “ s_k ” is denoted by $Encrypt_{s_k}$. The Miyaguchi–Preneel compression function is an iterated function that initially stores the hash values through the generator function “ f ”, that is to be converted to fit the size of the

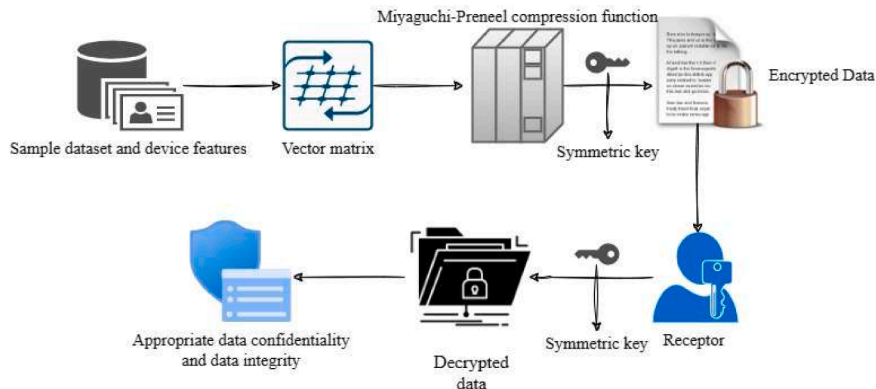


Fig. 2. Miyaguchi–Preneel compression function-based network model.

key of the cipher “ K_i ”, the previous hash value $Hash_{i-1}$ is every time passed to the function “ f ” to match the size for the key length of block cipher, and the output of the compression function is mathematically stated in Eq. (2):

$$Hash_i = Encrypt_{G(Hash_{i-1})}(M_i) \oplus Hash_{i-1} \oplus (M_i) \quad (2)$$

The Eq. (2) explains that the adopted Miyaguchi–Preneel compression function stores each block of the device data holding the plain text to be encrypted as a block formulating a vector matrix “ VM ”. The value of $i = 1, 2, 3, \dots, x$ and $Hash_i$ is the corresponding hash value. The hash values of two message blocks are not same $Hash_1 \neq Hash_2$. The plaintext here is denoted as M_i that stores the device data and the sample data used for simulation. The previous hash value $Hash_{i-1}$ is added to the device data forming the plaintext to be encoded. The resultant cipher text is then XORed with the message M_i and then again XORed with the previous hash value $Hash_{i-1}$ to generate a new hash value $Hash_i$. The preceding hash value $Hash_{i-1}$ is stored as the key for block. At the initial stage, where there is no preceding hash value, it uses persistent pre-determined hash value $Hash_0$ for block cipher. Here, the functioning of M_i and $Hash_{i-1}$ may be swapped accordingly in a way that After evaluating hash values for the plain text, the converted text is transmitted to the indented user that uses the symmetric key to retrieve the source or original text and thus the communication pattern between the devices is analysed to construct an individual device profile. Moreover, the data confidentiality and data integrity are maintained ensuring the secrecy and privacy of device data flow within a network.

The zero-trust network in the IIoT operates completely on devices without manual intervention. the involvement of devices in the network makes the network more prone to attacks whether intruders or external attacks. The device data held by the blocks in the underlying approach are protected against unauthorized access or malicious threats with the aid of explained compression function that ensures the confidentiality and originality of the data involved in the network. assigning hash values to the device data benefits in sustaining the uniqueness of the device data. The dominating factor of using the hash compression function is that, when a single modification is made in the source data, the initially assigned hash values also get changed automatically to address the modification. Thus, the compression function is witnessed as a sound mechanism that certifies and guarantees the confidentiality and integrity of the source data of the device ready for communicating in the network. then the substitution permutation set is performed on the assigned hash values to establish strong data confidentiality and data integrity by generating cipher text for the associated plain text.

5.3. Device profiling with the aid of substitution-permutation network

Substitution permutation mechanism is applied on the produced and assigned hash values to each block holding the corresponding plaintext for the device data. The substitution-permutation network takes the hash values generated for individual device data block and the private key Key_{pvt} as input for producing cipher text for the corresponding plain text by shuffling the bits through different random combinations from “Substitution boxes (Sub-Boxes)” and “Permutation boxes (Perm-Boxes)”. Substitution permutation works as mixing the bits and the arranging them as a new sequence of bits concatenating with the secret key. The overall functioning of substitution-permutation breakdown in three sub-processes- addition of secret key Key_{pvt} , then shuffling or substituting the bits, and at last performing the permutation of bits to generate a new sequence of bits that is considered as the cipher text for the associated source or plain text. The plaintext block is XORed with the associated secret key Key_{pvt} , which is assigned to each round in the SPN. Multiple

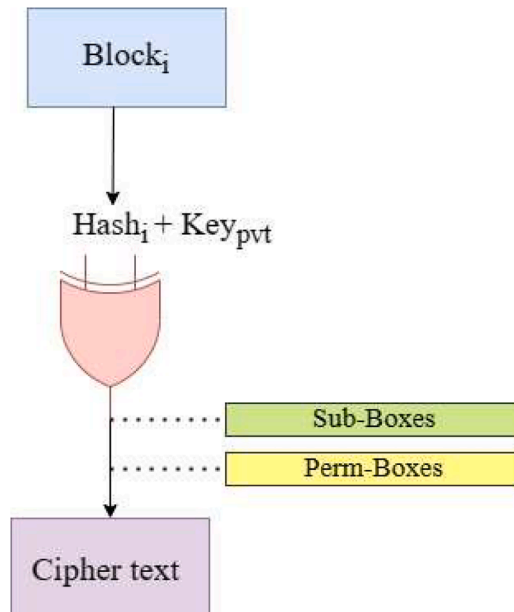


Fig. 3. structure of Miyaguchi–Preneel compression function-based substitution-permutation network.

rounds comprise of number of layers alternatively made of substitution permutation combinations. The substitution-permutation sequences in the network are made public over the network only the associated private keys are kept secret. The text over the network is arranged in blocks. The model and functioning of substitution-permutation network is demonstrated in the below Fig. 3.

The Fig. 3 explains the overview of the functioning of the Miyaguchi–Preneel compression function-based substitution-permutation network for obtaining the cipher text of the corresponding plain text and in turn constructs a behavioural profile based on the communication pattern and the security level of the resources required by the devices in the network. as illustrated in the figure, during each session of substitution-permutation sequence individual secret or private keys are added to analyse the communication pattern of the devices in the network to create a device profile. At the beginning of each round, the source data that is the data of the device is XORed with the secret key of the associated block to generate a device profile for all the devices in the IIoT environment. the employment of secret or private key ensures that the original plain text is retrieved from the cipher text with the intended or valid receptor device that holds the private key. The adding of secret keys to the source data is mathematically stated in Eq. (3):

$$R_{key} = \beta_i + Key_{pvt} \quad (3)$$

In Eq. (3), the adding of private key to the corresponding device data is performed and after rounding or adding the private key with the individual block, next, the bits of the original data are shuffled and substituted among themselves and a new sequence of substituted bits is created and then organised as blocks. the new arrangement of bits obtained is substituted with the aid of homogenous key for encryption. The substitution of bits is mathematically formulated in Eq. (4):

Algorithm 1

Input: Device data set $\beta = \{\beta_1, \beta_2, \beta_3 \dots \dots \beta_x\}$ and sample dataset involved in simulation $\alpha = \{\alpha_1, \alpha_2, \alpha_3 \dots \dots \alpha_y\}$

Output: Enhanced and certified data confidentiality and data integrity, hence authentic data communication

1. Initialize the value of “x”, “y”, $B_i = x$ and $K_i = 1$
 2. Begin
 3. For each device data and the sample data
 4. Formulate a vector matrix as modelled in equation (1).
 - \\ To encrypt the device data
 5. Perform Miyaguchi–Preneel compression function to produce hash value for each device data

$$Hash_i = Encrypt_{G(Hash_{i-1})}(M_i) \quad Hash_{i-1}(M_i)$$
 6. For each block with source device data, do
 7. Add respective round key or secret key Key_{pvt} to the associated plain text

$$R_{key} = \beta_i + Key_{pvt}$$
 8. Shuffle the bits among themselves to create a random sequence of bits using the symmetric key s_k , known as “Substitution”

$$Subs(\beta_i \rightarrow \beta_j) = s_k(\beta_i \rightarrow \beta_j)$$
 9. Combine the device data bits around leaving the first row, known as “Permutation”

$$Perm(\beta_i \rightarrow \beta_j) = Shift_{row}(\beta_i \rightarrow \beta_j)$$
 10. Corresponding cipher text for the associated plaintext
 - \\ To decrypt the encrypted device data, perform inverse of above procedure of substitution-permutation network
 11. Perform reverse permutation function

$$Perm''(\beta_i \rightarrow \beta_j) = Shift_{row}''(\beta_i \rightarrow \beta_j)$$
 12. Execute inverse of substitution function

$$Subs''(\beta_i \rightarrow \beta_j) = s_k''(\beta_i \rightarrow \beta_j)$$
 - \\ verification
 13. If $Perm''(\beta_i \rightarrow \beta_j) = Perm(\beta_i \rightarrow \beta_j)$ and $Subs''(\beta_i \rightarrow \beta_j) = Subs(\beta_i \rightarrow \beta_j)$, then
 14. Authenticity is certified and device is permitted to initiate the communication
 15. Else
 16. Authentication is not certified and device is blocked to initiate communication
 17. End if
 18. End for
 19. End
-

$$\text{Subs}(\beta_i \rightarrow \beta_j) = s_k(\beta_i \rightarrow \beta_j) \quad (4)$$

As discussed in Eq. (4), the substitution of hash values by shuffling of bits to generate new random sequence of bits is performed. After completing the substitution, permutation of bits is achieved where the device data bits are combined and all the rows in the vector-matrix are shifted leaving the first row. The permutation phase is mathematically represented as in Eq. (5):

$$\text{Perm}(\beta_i \rightarrow \beta_j) = \text{Shift}_{\text{row}}(\beta_i \rightarrow \beta_j) \quad (5)$$

The shifting of rows or the permutation of bits is mathematically represented in Eq. (5). Following the above-explained three phases of the substitution-permutation network, the required cipher text is obtained for the corresponding source device data and thus the two broad considerations data confidentiality and data integrity of the IIoT device data involved in the zero-trust network is certified. Moreover, the device profile created at this stage embraces the personalized authentication of the devices based on the usual behaviour or patterns associated with the individual device and the security level noted for each device resource requirement is established. The algorithm for this section of the overall proposed model is as illustrated below:

The Algorithm 1, explains the functioning of adopted the Miyaguchi–Preneel compression function-based substitution-permutation network (MPC-SP) to achieve improved data confidentiality and data originality to promote secure and trustworthy device connections and communication of IIoT environment over zero trust network. The MPC-SP comprises of individual steps that together ensures secured data communication. Foremost step in the process is to generate a source vector matrix that comprises of the data of the IIoT devices and the samples involved in simulation. the single device data is maintained in a individual block. Next, the hash value is calculated and then assigned to the plaintext or the source device data to maintain the originality of the data with the aid of Miyaguchi–Preneel compression function. Evaluating hash values for the source data ensures that the source data will not be accessed through any unauthorized or malicious device in the network. After assigning the hash values, the device data is concatenated with the associated secret key uniquely allotted to each device data block. This technique is termed as “rounding keys”. Then the bits of the device data are shuffled among themselves to generate a new sequence of bits called “substitution” and then the formed new sequence of random bits are mixed around except the first row. This mechanism of mixing bits is known as “permutation”. The substitution-permutation is initiated and accomplished with the aid of Substitution boxes “(Sub-Boxes)” and Permutation boxes “(Perm-Boxes)”. Following the above explained process, the encrypted data is achieved for the source device data or the plain text. The generated encrypted data is then transferred to the intended receptor device. The device then decodes or decrypt the encrypted data with the aid of same key used for encryption of device data. For decryption, the inverse of substitution-permutation process is executed with the aid of secret key associated with each block data during the rounding keys. At last, the decision making is processed to verify the authenticity of the device. The device is said to be authentic and granted access to explore and communicate over the network only when the inverse substitution-permutation values match the prior evaluated substitution-permutation values, contrary, in case of mismatching values, the device is claimed as unauthentic and the network access to the device is blocked.

The profiling of device is created during the beginning of each substitution-permutation cycle by examining the device communication patterns and the level of security required by the resources desired by underlying device. Device profiling is then observed and brought into practice to personally build a authorization criteria irrespective of going across multiple stages of deciding the authenticity and authorize the device to access the network environment.

5.4. Transfer fuzzy learning in evaluating trust values depending upon the security level of resources

Transfer fuzzy learning is employed with the aim of producing enhanced and more accurate learnings from the past acquired but related data or knowledge in practice. Here, the transfer fuzzy learning is deployed to improve and intensify the IIoT environment over

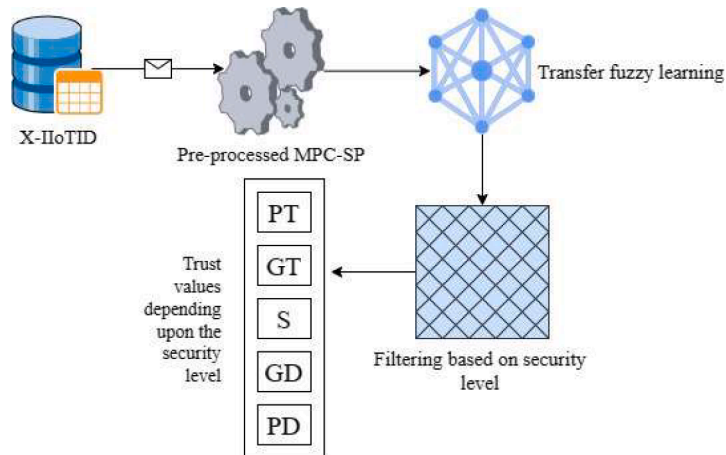


Fig. 4. Structure of transfer fuzzy learning in evaluating trust values depending upon the security level of resources.

zero-trust security model by obtaining the information produced by Miyaguchi–Preneel compression function-based substitution-permutation network domain. the concern of bringing transfer fuzzy learning in practice is to evaluate the requirement of security levels of the resources associated with the IIoT devices which in turn facilitates smooth and effective personalized device authentication. With the aid of transfer fuzzy learning, the trust values are calculated based on the level of security desired for the device resource. Here, the device with highest security is marked as a device with “pure trust”, device with less resource security level than purely trusted device is marked as “Generic trust”. Then falls the device with moderate security level required for resources is claimed as “Satisfactory” and then the trust value category with minimal and no security requirement is identified as “Generic distrust” and “Pure distrust” respectively. The basic model of transfer fuzzy learning in evaluating trust values depending upon the security level of resources is modelled in Fig. 4.

As illustrated in the Fig. 4, the transfer learning acquires information from pre-processed domain i.e.; Miyaguchi–Preneel compression function-based substitution-permutation network. The information is then passed to the fuzzy learning sub-system. The fuzzification process is initially performed where the transfer fuzzy learning is applied to pre-trained information to evaluate the trust values based on the requirement of security possess by individual device. The process of creating fuzzy set from the crisp values is denoted as the fuzzification and the generation of crisp values from the obtained fuzzy set is termed as defuzzification. This loop at last provide the values that empowers the decision making process in authenticating the devices. The filtering of devices is then made and

Algorithm 2

Input: Device data set $\beta = \{\beta_1, \beta_2, \beta_3 \dots \dots \beta_x\}$ and sample dataset involved in simulation $\alpha = \{\alpha_1, \alpha_2, \alpha_3 \dots \dots \alpha_y\}$

Output: Robust categorization of devices based on trust values

1. Initialize the pre-trained information from MPC-SP.
 2. Begin
 3. For all inputs of device data βI_i and their corresponding outputs βO_j
 4. Formulate a source domain

$$D_{src} = (\beta I_i, \beta O_j, \forall T \in \alpha)$$
 5. Evaluate source domain interval for the two input devices

$$D_{src}I = \{(\beta I_1, \beta I_2), (\beta O_1, \beta O_2)\}$$
 6. For the two devices, do
 7. Examine the device and its resource and calculate the trust “T” of device on its associated resource to determine the level of security

$$T_k = \frac{Avg(T_a)}{1 - [(Avg(T_a, R_a)) + (Avg(T_b, R_b))]}$$
 8. If ‘ $5 > T \geq 4$ ’, then
 The device and resource establish “Pure Trust”
 9. End if
 10. If ‘ $4 > T \geq 3$ ’, then
 The device and resource establish “Generic Trust”
 11. End if
 12. If ‘ $3 > T \geq 2$ ’ then,
 The relationship between the device and resource is “Satisfactory”
 13. End if
 14. If ‘ $2 > T \geq 1$ ’, then
 The device and resource establish “Generic Distrust”
 15. End if
 16. If ‘ $1 > T \geq 0$ ’, then
 The device and resource establish “Pure Distrust”
 17. End if
 18. End for
 19. End for
 20. End
-

accordingly the categorization is performed labelling the trust values of the devices ranging from highest to the least or no security as: “Pure Trust”, “Generic Trust”, “Satisfactory”, “Generic Distrust”, “Pure Distrust” respectively. The trust values obtained are used to indicate different level of security used to distinguish the devices and hence make personalized authentication depending on the fuzzy set. The transfer learning source domain is mathematically stated as in Eq. (6):

$$D_{src} = (\beta I_i, \beta O_j, \forall T \in \alpha) \quad (6)$$

The Eq. (6) initially formulates source domain based on the number of inputs of device data represented as βI , the outputs of individual device data (βO) and all the rows of the vector matrix for the sample “ α ”. Based on the source domain for transfer learning, the next is to evaluate the domain source interval between the devices. The Eq. (7) represents the mathematical modelling of domain interval:

$$D_{src}I = \{(\beta I_1, \beta I_2), (\beta O_1, \beta O_2)\} \quad (7)$$

As discussed in Eq. (7), after formulating a source domain of the pre-trained input and output, the source domain interval “ $D_{src}I$ ” is evaluated based on the two device data inputs and their corresponding outputs ($\beta I_1, \beta I_2$) and ($\beta O_1, \beta O_2$) respectively. Depending on the interval between two inputs and their outputs, the final set of source domain input output is generated for all the participating device data inputs and their outputs represented as x and y respectively. The Eq. (8) states the mathematical representation of source domain:

$$D_{src} = \sum \beta I_i, \sum \beta O_j \quad (8)$$

The Eq. (8) explains the summarization of all the device data inputs βI_x and device outputs βO_y to create a source domain for all x and y respectively. After formulating the source domain based on the pre-trained data, the trust value between the device and the resource to be accessed is calculated using Eq. (9):

$$T_k = \frac{\text{Avg}(T_a)}{1 - [(\text{Avg}(T_a, R_a)) + (\text{Avg}(T_b, R_b))]} \quad (9)$$

From the Eq. (9), the security level required for the resource to access the device is analysed and then based on the intensity of security the trust value is evaluated to add the information to the profile created by the Miyaguchi–Preneel compression function-based substitution-permutation network for efficient and appropriate personalized authentication of devices in zero trust network. Trust “ T ” is evaluated by the average trust on the resources of the device representing the trust T_a of device “ β_a ” on resource R_a and the trust T_b of device “ β_b ” on resource R_b . After evaluating trust, the input fuzzy set for trust values is shown in Eq. (10):

$$F_s \beta I_i = \{PT, FT, S, GD, PD\} \quad (10)$$

The Eq. (10) above demonstrates the fuzzy input sets for all “ x ” and is represented as $F_s \beta I_x$. The fuzzy set is generated to reflect the security level of the resources desired by individual device in the network. The trust values are classified not only as “trust” and “distrust”, but have the values in the scale labelling as “Pure Trust”, Generic Trust, Satisfactory, Generic Distrust and Pure Distrust, identified as PT, GT, S, GD and PD respectively. The different trust values indicate that not all the devices in the network have same requirement of resources of same security level. The step-by-step pseudo code of personalized device authentication with the aid of transfer fuzzy learning is as follows:

The Algorithm 2, explains the functioning of transfer fuzzy learning-based trust value evaluation to identify the level of resources a particular device desires to access. The foremost step is to formulate a source domain with all the input device data and their corresponding outputs. With the generated source domain, now the domain interval is calculated between the devices. Next, not all the devices are going to access the same level of security for the desired resources. Therefore, the trust is calculated, which actually determines the level of security required by individual device for accessing different associated resources. Based on the calculated trust,

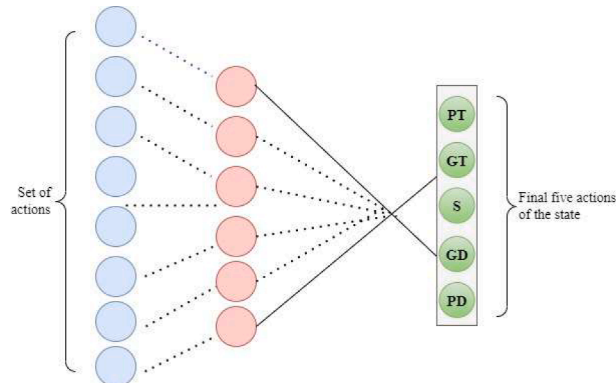


Fig. 5. Structure of choosing action by an agent.

input fuzzy set is constructed splitting the devices under parameters “Pure Trust”, Generic Trust, Satisfactory, Generic Distrust and Pure Distrust, identified as PT, GT, S, GD and PD respectively that provides an ambient solution to personalised device authentication based on the profile created and the level of security desired is achieved.

5.5. Reinforcement learning based decision making for personalised authentication (RL-PA)

Reinforcement learning is a sub category of machine learning that employees the “agent” in the environment comprising of overall functioning of underlying system to perform certain action and accomplish the requested associated state or task. The intelligent agent ought to analyse and examine the state of the environment, here, the IIoT environment, and accordingly activates to perform the best possible action with the aim to maximize the its rewards based on its actions. The basic representation of intelligent agent state-action loop is demonstrated in Fig. 5.

The Fig. 5 demonstrates that the agent performing in a particular environment, on receiving a state, went through number of actions to accomplish the specified task. Here, the agent performs set of actions and at last provide the action outcome from the set of evaluated fuzzy values only. Therefore, appropriate and sound decision making is achieved by the agent in determining the device and facilitate the zero-trust network with personalised device authentication. If the value of action lies under the mentioned five categories, then only the device is considered as the member of network and even with poor trust or weak security, is granted permission to access the network otherwise, the device with unusual profile is blocked and warned as “malicious”.

The decision making is facilitated by the reinforcement learning by employing an intelligent agent that keeps track on the states entering the environment and thus performs set of different actions to reach a final action to determine the level of security for accessing different resources. The agent based on its action for the particular state receives reward from the environment. the reward here, indicates the accuracy in choosing the best optimised action for the particular state the agent is going through. The overall functioning of agent in an IIoT environment is summarized in Fig. 6.

As represented in the Fig. 6. the agent is present in the environment and at the arrival of a particular state, the agent gets active and start performing set of actions to finally achieve the best and optimised action from the “action set”. The actions here are divided into five desired categories listing as – “PT”, “GT”, “S”, “GD”, “PD” that falls under actions a_1 , a_2 , a_3 , a_4 , and, a_5 respectively and hence ensure personalised device authentication thereafter. The agent chooses the most optimal action from the action set “A”. The mentioned actions are “PT”, “GT”, “S”, “GD”, “PD” denoting “Pure trust”, “Generic trust”, “Satisfactory”, “Generic distrust” and “Pure distrust” respectively. In decision making, the agent while performing actions aims to maximize the rewards it receives from the environment also known as “external reward” rather than the reward agent receives from an intrinsic function based on the current state in which the agent is operating. The overall reinforcement learning breaks down into three major sections namely: 1. Action, 2. Rewards and 3. decision making. “Action” is performed by the agent to choose the most appropriate step based on the state arrived. Depending on the action taken by the agent, the rewards are then assigned to motivate the action of the agent. Lastly, the decision is made to verify the authenticity of the device in the network.

Action: To accomplish a particular specified task in a given state, agent “A” went through number of actions to reach a desired and optimised action. The set of actions is denoted as $a_i(s)$ that is applied to a particular state $s \in S$ and the action $a_i(s) \in A$, where $i \rightarrow$ “PT”, “GT”, “S”, “GD”, “PD”

Rewards: Rewards can be explained as the values assigned to the agent based on the accuracy in achieving the desired and optimised action from the set of actions for the particular state. The rewards agent received are categorized as immediate reward and cumulative reward. The immediate rewards are the one agent receives from the environment based on its performance also known as external reward Rw_t^E . The rewards that are assigned to the agent internally to encourage its performance using the internal function is termed as “internal or immediate reward” Rw_t^I . In RL, the agent’s performance is delineated based on the reward passed from environment to agent.

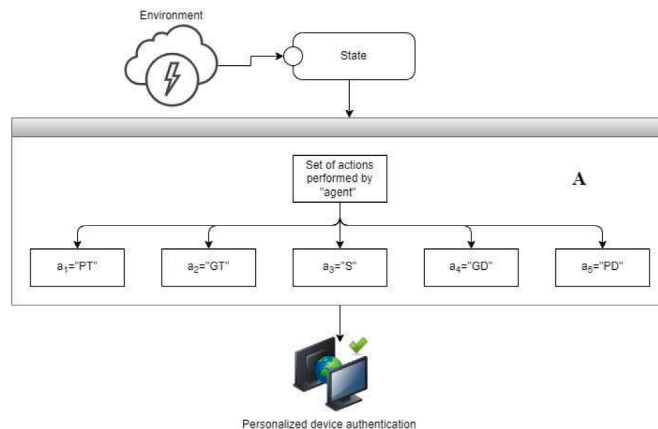


Fig. 6. The overall functioning of agent in performing and achieving optimised action.

Decision making: in RL, the decision for the environment state is taken by the agent based on the five mentioned action values among the set of action. the agent marks the device based on the security requirements denoted as "PT", "GT", "S", "GD", "PD" representing the action values a_1 , a_2 , a_3 , a_4 , and, a_5 respectively. The pseudocode explaining step by step working of RL-PA is delineated below:

As illustrated in the Algorithm 3, the agent is trained intelligently to analyse the entered state of the environment and to choose the best optimized action from the fuzzy set constructed on the basis of different security requirements of the devices to access the resources in the network. at the initial stage, the agent is activated as the state arrives in the environment the agent is operating. Then the agent performs action a_T for state s_T arriving at time T. the intelligent agent chooses best optimized action from the fuzzy set created "PT", "GT", "S", "GD", "PD". The actions are categorized into 5 categories a_1 , a_2 , a_3 , a_4 , and, a_5 denoting the different trust values respectively. Depending on the calculated action from the agent, the decision is made by the system based on the device profile and the trust values that denote the security level required for resource access. based on the efficiency and accuracy of agent in selecting the most relevant and optimized action, the rewards are passed to the agent from the environment and from the internal function based on the current state denoted as Rw_T^E and Rw_T^I with the aid of external reward generator function and internal reward generator function denoted as f_E and f_I respectively. The overall reward Rw_T is calculated by adding both the internal and external rewards.

The Algorithm 4 delineates the overall working of proposed PDA-QLTFL. Initially, the compression function is performed to assign hash values to the device data. Next, the encryption/decryption is performed with the aid of substitution permutation networks which in return helps in constructing the profile of the device by reading the communication patterns of the devices. then the trust value is evaluated for each device to verify its authenticity and accordingly the action is performed by the agent to get reward. The agent aims to maximize the rewards it receives. Thus, the personalized device authentication is achieved with the aid of device profiling without compromising the data confidentiality and integrity with more optimized and accurate decision-making via reinforcement learning.

Algorithm 3

Input: the input fuzzy set, agent A, action "a, state "s", environment E, cumulative reward Rw_T^E , immediate reward Rw_T^I , time T, state generator function f_s , external reward generator function f_E , internal state generator f_I

Output: optimised decision making for personalised device authentication

1. Initialize A and state generator function f_s
 2. Assign action $\rightarrow a$ and state $\rightarrow s$ at each time interval T
 3. Begin
 4. For all state "s", do
 5. Evaluate a_i at time T, $\forall a_i \in \text{"PT", "GT", "S", "GD", "PD"}$
 6. If $a_i = \text{"2"}$, do
 7. Mark the device "pure trust" with resource then calculate overall reward:

$$Rw_T = Rw_T^E + Rw_T^I$$
 8. End if
 9. If $a_i = \text{"1"}$, do
 10. Mark the device "generic trust" with resource then calculate overall reward:

$$Rw_T = Rw_T^E + Rw_T^I$$
 11. End if
 12. If $a_i = \text{"0"}$, do
 13. Mark the device "Satisfactory" with resource then calculate overall reward:

$$Rw_T = Rw_T^E + Rw_T^I$$
 14. End if
 15. If $a_i = \text{"-1"}$, do
 16. Mark the device "generic distrust" with resource then calculate overall reward:

$$Rw_T = Rw_T^E + Rw_T^I$$
 17. End if
 18. If $a_i = \text{"-2"}$, do
 19. Mark the device "pure distrust" with resource then calculate overall reward:

$$Rw_T = Rw_T^E + Rw_T^I$$
 20. End if
 21. End for
-

Algorithm 4

Input: Device data set $\beta = \{\beta_1, \beta_2, \beta_3 \dots \dots \beta_x\}$ and sample dataset involved in simulation $\alpha = \{\alpha_1, \alpha_2, \alpha_3 \dots \dots \alpha_y\}$, the input fuzzy set $F_s \beta I_i$, agent A, action "a, state "s", environment E, cumulative reward Rw_T^E , immediate reward Rw_T^I , time T, state generator function f_s , external reward generator function f_E , internal state generator f_I

Output: Personalized device authentication with improved data confidentiality rate, better data integrity rate, reduced false positive rate and minimal network access time

1. Initialize the value of "x", "y", $B_i = x$ and $K_i = 1$
2. Begin
3. For each device data and the sample data
4. Formulate a vector matrix as modelled in equation (1).

\\ To encrypt the device data

5. Perform Miyaguchi–Preneel compression function to produce hash value for each device data

$$Hash_i = Encrypt_{G(Hash_{i-1})}(M_i) Hash_{i-1}(M_i)$$
6. For each block with source device data, do
7. Add respective round key or secret key Key_{pvt} to the associated plain text

$$R_{key} = \beta_i + Key_{pvt}$$
8. Shuffle the bits among themselves to create a random sequence of bits using the symmetric key s_k , known as "Substitution"

$$Subs(\beta_i \rightarrow \beta_j) = s_k(\beta_i \rightarrow \beta_j)$$
9. Combine the device data bits around leaving the first row, known as "Permutation"

$$Perm(\beta_i \rightarrow \beta_j) = Shift_{row}(\beta_i \rightarrow \beta_j)$$
10. Corresponding cipher text for the associated plaintext

\\ To decrypt the encrypted device data, perform inverse of above procedure of substitution-permutation network

11. Perform reverse permutation function

$$Perm''(\beta_i \rightarrow \beta_j) = Shift_{row}''(\beta_i \rightarrow \beta_j)$$
12. Execute inverse of substitution function

$$Subs''(\beta_i \rightarrow \beta_j) = s_k''(\beta_i \rightarrow \beta_j)$$
13. If $Perm''(\beta_i \rightarrow \beta_j) = Perm(\beta_i \rightarrow \beta_j)$ and $Subs''(\beta_i \rightarrow \beta_j) = Subs(\beta_i \rightarrow \beta_j)$, then
14. Authenticity is certified and device is permitted to initiate the communication
15. Else
16. Authentication is not certified and device is blocked to initiate communication
17. End if

18. For all authentic devices, do

19. Initialize the pre-trained information from MPC-SP.

20. Begin

21. For all inputs of device data βI_i and their corresponding outputs βO_j

22. Formulate a source domain

$$D_{src} = (\beta I_i, \beta O_j, \forall T \in \alpha)$$

23. Evaluate source domain interval for the two input devices

$$D_{src}I = \{(\beta I_1, \beta I_2), (\beta O_1, \beta O_2)\}$$

24. For the two devices, do

25. Examine the device and its resource and calculate the trust "T" of device on its associated resource to determine the level of security

(continued on next page)

Algorithm 4 (continued)

$$T_k = \frac{Avg(T_a)}{1 - [(Avg(T_a, R_a)) + (Avg(T_b, R_b))]}$$

26. If ' $5 > T \geq 4$ ', then
The device and resource establish "Pure Trust"
27. End if
28. If ' $4 > T \geq 3$ ', then
The device and resource establish "Generic Trust"
29. End if
30. If ' $3 > T \geq 2$ ' then,
The relationship between the device and resource is "Satisfactory"
31. End if
32. If ' $2 > T \geq 1$ ', then
The device and resource establish "Generic Distrust"
33. End if
34. If ' $1 > T \geq 0$ ', then
The device and resource establish "Pure Distrust"
35. End if
36. For all entering state art time T
22. Evaluate a_i at time T, $\forall a_i \in \text{"PT", "GT", "S", "GD", "PD"}$
23. If $a_i = \text{"2"}$, do
24. Mark the device "pure trust" with resource then calculate overall reward:
 $Rw_T = Rw_T^E + Rw_T^I$
25. End if
26. If $a_i = \text{"1"}$, do
27. Mark the device "generic trust" with resource then calculate overall reward:
 $Rw_T = Rw_T^E + Rw_T^I$
28. End if
29. If $a_i = \text{"0"}$, do
30. Mark the device "Satisfactory" with resource then calculate overall reward:
 $Rw_T = Rw_T^E + Rw_T^I$
31. End if
32. If $a_i = \text{"-1"}$, do
33. Mark the device "generic distrust" with resource then calculate overall reward:
 $Rw_T = Rw_T^E + Rw_T^I$
34. End if
35. If $a_i = \text{"-2"}$, do
36. Mark the device "pure distrust" with resource then calculate overall reward:
 $Rw_T = Rw_T^E + Rw_T^I$
37. End if
38. End for
39. End for
40. End for
41. End for
42. End for
43. End

6. Experimentation, results and analysis**6.1. Experimental setup**

The experimentation performed in this section explains the comparison between the adopted approach Personalized device authentication scheme using Q-learning-based decision-making with the aid of transfer fuzzy learning for IIoT devices in zero trust

network (PDA-QLTFL) and the peer approaches Consistent Round Hash optimized SRP-6a-based end-to-end mutual authentication for secure data transfer in industry 4.0 [1] and The Cyber Security via Determinism", Paradigm for a Quantum Safe Zero Trust Deterministic Internet of Things (IoT) [2]. The experiment is conducted using Python on the simulation dataset extracted from X-IIoTID dataset. The parameters on which comparison and experiment is performed are the false positive rate, data confidentiality rate, data integrity rate and the network access time. The comparison of PDA-QLTFL to other existing reinforcement learning based device authentication is illustrated in Table 1.

The experimental comparative analysis of the proposed approach and the existing approaches is demonstrated in Table 2.

6.2. Result analysis

This section explains the results obtained from proposed PDA-QLTFL method empowered zero trust network and the peer approaches. The performance evaluation is made on the following parameters:

- False positive rate
- Data confidentiality rate
- Data integrity rate
- Network access time

The tables and graphs below illustrate the performance of the proposed and existing approaches.

6.2.1. False positive rate

False positive rate is measured as the ratio of untrue events that are erroneously categorised as correct or true event. To ensure minimal false positive rate in the zero-trust network is the key consideration to be looked after. The mathematical statement for evaluating false positive rate is stated in Eq. (11):

$$fpr = \frac{fp}{fp + tn} * 100 \quad (11)$$

The Eq. (11) illustrates that the false positive rate is evaluated on two variables false positive and true negative represented as fp and tn respectively. The ratio is obtained using the two variables to retrieve the false positive rate. The proposed approach (PDA-QLTFL) produces better results with minimal false positive rate as compared to the existing approaches Consistent Round Hash optimized SRP-6a-based end-to-end mutual authentication for secure data transfer in industry 4.0 [1] and The Cyber Security via Determinism", Paradigm for a Quantum Safe Zero Trust Deterministic Internet of Things (IoT). The comparative table of the three approaches is as follows:

In Table 3 and Fig. 7, the graphical representation of the comparison results is performed on sample involved in simulation process ranging from 1000 to 100,000. The graph explains the performance evaluation with x-axis denoting the sample dataset and y-axis representing the outcomes of the false positive rate. The results obtained state that the proposed approach PDA-QLTFL has better performance with reduced false positive rate as compared to the [1] and [2] respectively. With the employment of PDA-QLTFL, the false positive rate is lessened by 3.182 % and 5.862 % as shown in Fig. 7 when compared with [1] and [2].

6.2.2. Data confidentiality rate

When transferring data in a IIoT environment, data confidentiality is the major aspect for any system for its efficient working and trustworthiness. Data confidentiality rate is explained as a rate at which the intended receptor receives the original data in transit without breach. IIoT environment over zero trust network builds on devices and to provide secure and leak resistant data transfer, data

Table 1
Comparison of different reinforcement learning methods.

Comparison parameters	Algorithm		
	Q-learning	SARSA	Deep Q-network (DQN)
Learning type	Off policy approach	On policy approach	Model-free, off-policy approach
Input	State-action pairs	Current state	Raw state
Output	Q-values for each state-action	Q-values for each state-action	Q value for each action
Convergent	Converges to a global solution assuming that after the generation of training and observation, the system transfers to the greedy policy.	Converges to a best solution assuming that the system keeps following the same policy that was followed at the time of obtaining experience	Speedy convergence
Application cases	Suitable in the cases where the performance of agent is not considered at the time of training	Suitable in the cases where agents' performance is considered at the time of training and gaining experience	Suitable in solving high dimensionality issues
Generalization	Limited to the states present in Q-table	Limited number of states	Not such limitations on states
Training time	Fast	Slow	Slow

Table 2

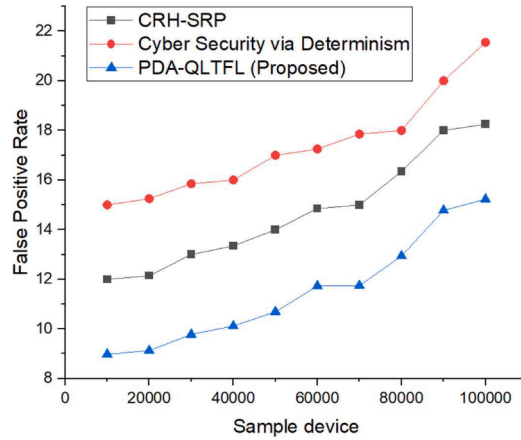
Comparative of different reinforcement learning methods in IIoT.

Approach	Accuracy	Security	False Positive Rate
DRL-IDS	90 %	x	x
SPC-DRRL	93.87 %	x	x
CNN-GRU	99.78 %	x	x
Proposed (PDA-QLTFL)	x	86.611	11.513

Table 3

Comparison results of false positive rate.

Sample device	CRH-SRP	Cyber Security via Determinism	PDA-QLTFL (Proposed)
10,000	12	15	8.98
20,000	12.15	15.25	9.13
30,000	13	15.85	9.78
40,000	13.35	16	10.12
50,000	14	17	10.69
60,000	14.85	17.25	11.73
70,000	15	17.85	11.75
80,000	16.35	18	12.94
90,000	18	20	14.78
100,000	18.25	21.55	15.23

**Fig. 7.** Sample device Vs False positive rate.

confidentiality must be ensured. Data confidentiality rate is measured as the amount at which the original data is received by the authorized device. The formula for calculating data confidentiality rate is given in Eq. (12):

$$C_R = \sum_{i=1}^y \frac{\alpha_{in*}}{\alpha_i} 100 \quad (12)$$

Table 4

Comparison results of data confidentiality rate.

Sample device	CRH-SRP	Cyber Security via Determinism	PDA-QLTFL (Proposed)
10,000	96.45	95.25	98.48
20,000	94.15	93.25	96.27
30,000	92	90	94.3
40,000	90	86.35	92.09
50,000	88.15	84	90.19
60,000	85	80	87.05
70,000	82	78.15	84.13
80,000	78	75	80.15
90,000	75.15	72	77.38
100,000	73	70	75.12

The Eq. (12) explains that the data confidentiality rate is calculated with the amount of data an intended receptor receives α_{in} to the total count of the sample devices α_i used for the simulation process. The table and graph below illustrate the comparative analysis of the proposed PDA-QLTFL and the referred approaches CRH-SRP and Cyber Security via Determinism.

The above-mentioned Table 4 and Fig. 8, explains the performance evaluation of the adopted approaches and the peer approaches. The result analysis provides that the PDA-QLTFL ensures better and enhance data confidentiality as compared to CRH-SRP and Cyber Security via Determinism. In graph, the x-axis denotes the sample dataset ranging from 1000 to 100,000 and y-axis represents the results of the data confidentiality rate. Moreover, PDA-QLTFL provides better data confidentiality rate with 2.126 % and 5.116 % as compared to [1] and [2] respectively.

6.2.3. Data integrity rate

The other key parameter that should be considered in IIoT environment over zero trust security network is the data integrity rate. Data integrity rate is understood as the device count that are not altered by any maliciously infected device to the total devices that are involved in the experimentation and performance evaluation. The formula for calculating the data integrity rate is stated in Eq. (13):

$$I_R = \sum_{i=1}^y \frac{\alpha_{original}}{\alpha_i} * 100 \quad (13)$$

The Eq. (13) above delineate that the data integrate rate is measured as the rate of number of devices that are not changed or intentionally altered by any attacker or unauthorized device in the form of user $\alpha_{original}$ to the total devices that are used as the sample for conducting the simulation α_i . the comparative analysis among the three approaches: PDA-QLTFL, CRH-SRP and Cyber Security via Determinism is presented in below table and graph.

The experiment is performed on the simulation dataset ranging from 10,000 to 100,000 plotted on the graph representing the x-axis illustrated in Table 5. The y-axis of the graph denotes the data integrity rate obtained on various data samples on the proposed approach PDA-QLTFL and the existing approaches CRH-SRP and Cyber Security via Determinism respectively. The adopted approach ensures enhanced rate of data originality and provides superior data integrity rate with 2.566 % and 5.446 % as compared to [1] and [2] respectively as shown in Fig. 9.

6.2.4. Network access time

Moreover, examining the three forementioned parameters, the next parameter under consideration is the network access time in a zero-trust security network. the network access is evaluated as the instant at which the associated network device makes a request to initiate a communication and the time frame at which the requested receptor device acquires the relevant request response resulting in a secure communication. The network access time is mathematically stated as in Eq. (13):

$$nat = \sum_{i=1}^y \alpha_i^{begin} - \alpha_i^{end} \quad (14)$$

The Eq. (14) above delineates the calculation of network access time “nat” which is based on the time a device takes in establishing a communication with the other device in the network. it is calculated as the time a device takes in putting a call to a device making a request α_i^{begin} to the time interval a device takes in responding to the request made and transferring that to the intended destination device α_i^{end} . The performance evaluation of the proposed technique PDA-QLTFL and the peer methods CRH-SRP and Cyber Security via Determinism is formulated in the table and graph below:

The comparative analysis is made on the three approaches with the sample dataset of size 10,000 to 100,000 formulated as in Table 6. The Fig. 10, explains that the x-axis denotes sample size and the y-axis represents the resultant network access time from the three approaches, PDA-QLTFL, CRH-SRP and Cyber Security via Determinism respectively. The analysis found reduces network access time by 1.25 % and 3.308 % as compared to the existing approaches

The proposed novel approach PDA-QLTFL found beneficial in providing a personalized device authentication thus empowering the IIoT system with sound security solution to the increased security requirements desired by the complex networks nowadays by restricting the continuous monitoring and verification. The approach enhances the data confidentiality and the data integrity thus resulting in a more secure and trustworthy IIoT environment over zero trust network. moreover, reduced false positive rate with less request response time in establishing a communication is facilitated by PDA-QLTFL, and therefore the research gaps in the existing approaches CRH-SRP and Cyber Security via Determinism is met by the novel approach resulting in a more efficient and productive IIoT infrastructure. the overall analysis if the research made so far emphasis on reduced false positive rate by 3.182 % and 5.862 % as compared to [1] and [2], better and enhanced data confidentiality rate by 2.126 % and 5.116 % as compared to [1] and [2], amplified data integrity with 2.566 % and 5.446 % when compared to CRH-SRP and Cyber Security via Determinism and minimal network access time by 1.25 % and 3.308 % as compared to [1] and [2] respectively.

7. Conclusion and future scope

The development of more complex networks, requires strong security requirements that are not met by the traditional security solutions. The zero-trust security model is the recent advancement in the security requirement that the complex networks desires. In the IIoT environment, where everything is managed and maintained by and on devices, the data is always prone to attacks by malicious

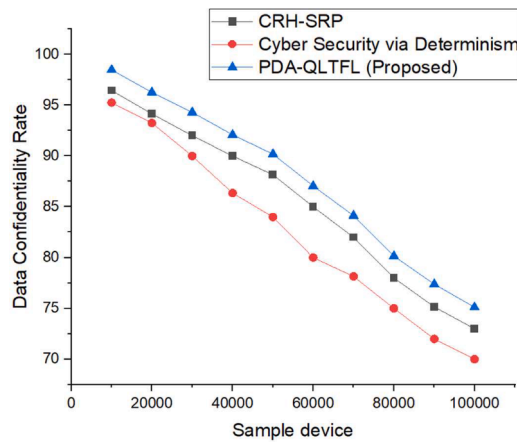


Fig. 8. Sample device Vs Data Confidentiality Rate.

Table 5

Comparison results of data integrity rate.

Sample device	CRH-SRP	Cyber Security via Determinism	PDA-QLTFL (Proposed)
10,000	97.75	94.15	99.84
20,000	94	92	96.29
30,000	92.15	90	94.74
40,000	89	85.15	91.69
50,000	85.35	82	88.24
60,000	82	79	84.79
70,000	78.15	75.15	80.64
80,000	75	73	77.79
90,000	70	68	72.57
100,000	68	64.15	70.47

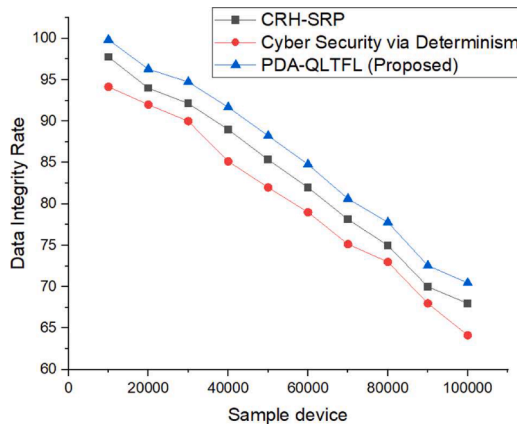


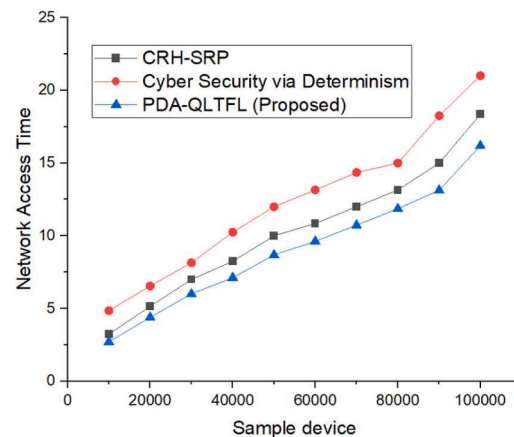
Fig. 9. Sample device Vs Data Integrity Rate.

devices during communicating over the network. moreover, the zero-trust security model, despite of number of improvements, consumes large amount of device energy in continuous verification every time the device seeks permission to access the network in IIoT environment. this hinders the productivity of the devices leaving the devices dry on monitoring and ensuring authenticity. The forementioned problems are all discussed and a novel solution is provided by the approach PDA-QLTFL in the paper. The approach offers better data confidentiality rate, enhanced data integrity rate, limiting the false positive rate and the network access time as well with 5.116 %, 5.446 %, 5.82 % and 3.308 % respectively as compared to the existing proposals. The approach integrates compression function to calculate hash values for each device data, then encryption and decryption is performed via substitution permutation networks. Next, the fuzzy set is generated based on the different security level required by the devices to access the resources and then the optimised decision is made with the aid of reinforcement learning at last. The security is the global concern and zero trust security

Table 6

Comparison results of network access time.

Sample device	CRH-SRP	Cyber Security via Determinism	PDA-QLTFL (Proposed)
10,000	3.25	4.85	2.71
20,000	5.15	6.55	4.39
30,000	7	8.15	6.01
40,000	8.25	10.25	7.127
50,000	10	12	8.69
60,000	10.85	13.15	9.62
70,000	12	14.35	10.73
80,000	13.15	15	11.86
90,000	15	18.25	13.14
100,000	18.35	21	16.19

**Fig. 10.** Sample device Vs Network Access Time.

witnessed as a ambient solution irrespective of the domain and operating environment. therefore, integrating artificial intelligence in zero trust network embraces data confidentiality and data originality to a great extent and therefore, zero trust with artificial intelligence will have a remarkable future in various sectors with complex networks demanding intelligent and strong security and decision-making infrastructure.

CRediT authorship contribution statement

Anamika Singh: Conceptualization, Investigation, Methodology, Writing – original draft. **Rajesh Kumar Dhanaraj:** Writing – review & editing, Project administration. **Anupam Kumar Sharma:** Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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